Y-Axis Actuator Sizing

The Y-axis of the Cartesian robotic arm is actuated by an encoder-equipped DC motor operating under closed-loop PID control, driving a timing belt and gantry cart mechanism. This setup translates the entire Z-axis assembly linearly along the Y-axis workspace, allowing the suction mechanism to reach and pick up target parts with high positional accuracy.

Governing Formulas

1. Torque Equations

- Total Motor Torque:

$$T_m = T_r + J_{\text{eq}} \cdot \ddot{\theta}_m$$

- Peak Required Motor Torque:

$$T_{m,\text{peak}} = \max\left(T_{m,\text{acc}}, T_{m,\text{const}}, T_{m,\text{dec}}\right)$$

- RMS Torque (T_{rms}):

$$T_{\text{rms}} = \sqrt{\frac{T_{m,\text{acc}}^2 \cdot t_{\text{acc}} + T_{m,\text{const}}^2 \cdot t_{\text{const}} + T_{m,\text{dec}}^2 \cdot t_{\text{dec}} + T_{m,\text{dw}}^2 \cdot t_{\text{dw}}}{t_{\text{acc}} + t_{\text{const}} + t_{\text{dec}} + t_{\text{dw}}}}$$

2. Inertia Equations

- Load Rotary Inertia (J_{load}):

$$J_{\text{load}} = \frac{m_{\text{load}} r^2}{\eta}$$

- Motor Rotary Inertia (J_{motor}):

$$J_{\text{motor}} = \frac{1}{10} \cdot J_{\text{load}}$$

- Equivalent System Inertia (J_{eq}):

$$J_{\text{eq}} = J_{\text{load}} + J_{\text{motor}}$$

3. Resisting Torque Equations

- Total Resisting Torque (T_r):

$$T_r = \frac{m_{\text{load}} r}{\eta}$$

4. Acceleration Equations

- Maximum Angular Velocity (ω_{max}):

$$v_{max} = \omega_{max} \cdot r$$

- Total Rotation, Area Under the Curve (θ):

$$\theta = \frac{1}{2} t_{acc} \omega_{max} + \omega_{max} t_{const} + \frac{1}{2} t_{dec} \omega_{max}$$

- Acceleration Phase ($\ddot{\theta}_{acc}$):

$$\ddot{\theta}_{acc} = \frac{\omega_{max} - 0}{t_{acc}}$$

- Constant-Speed Phase ($\ddot{\theta}_{const}$):

$$\ddot{\theta}_{const} = 0$$

- Deceleration Phase ($\ddot{\theta}_{dec}$):

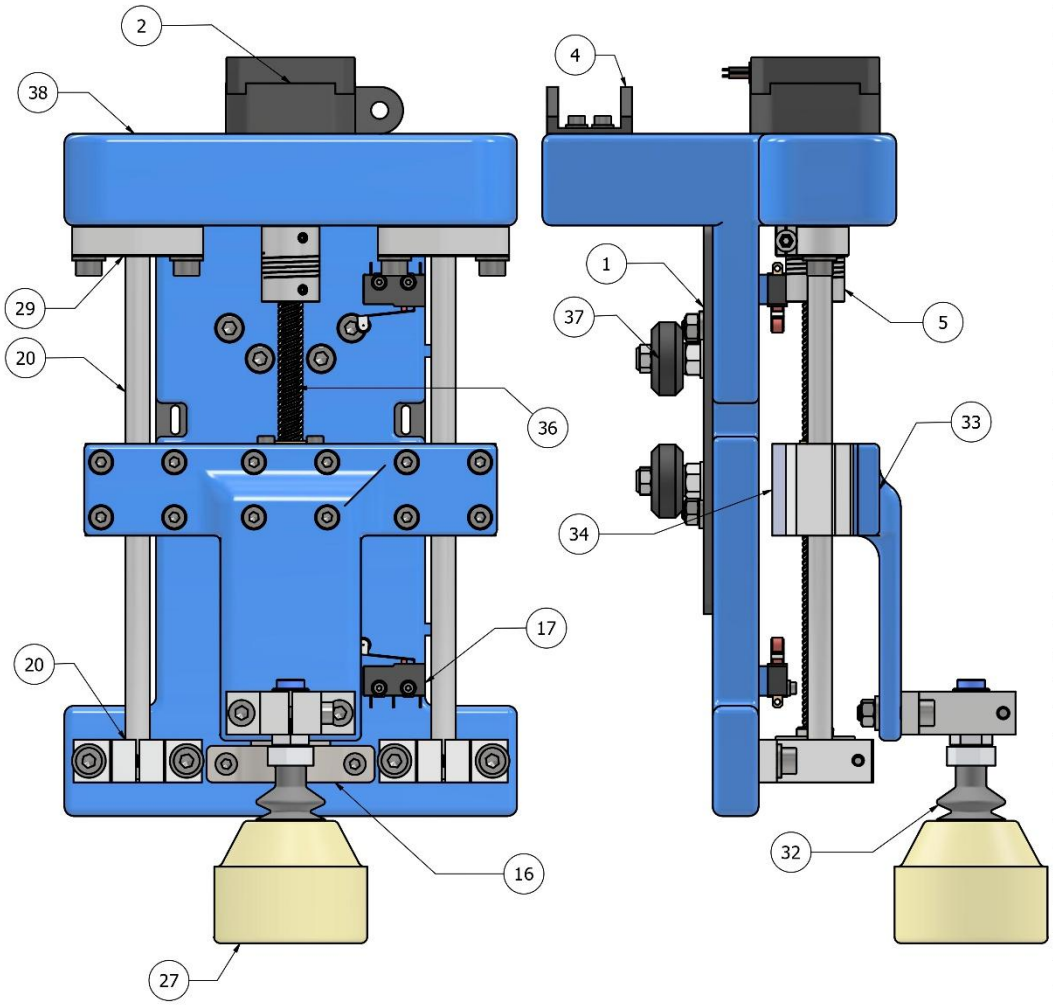
$$\ddot{\theta}_{dec} = \frac{0 - \omega_{max}}{t_{dec}}$$

5. Additional Rules and Parameters

- Driver Pulley
  - Number of Teeth $T = 30$
  - Pitch $P = 2\ mm$
  - Pitch Radius (r):

$$r = \frac{T \times P}{2\pi} = \frac{30 \times 2}{2\pi} = 9.55\ mm$$

To size the actuator correctly, the total mass moved along the Y-axis is determined first. Summing all translating parts in the Z-Axis Parts List gives a total mass of **1,363.87 g**.



Z-Axis			
No.	Qty.	Part Name	Mass
1	1	127x88 Gantry Plate	48.00 g
2	1	17HS4401 Stepper	280.00 g
3	1	Bearing Spacer	1.53 g
4	1	Drag Chain Link	1.80 g
5	1	Flexible Shaft Coupling	16.00 g
6	4	ISO 4032 - M2	0.48 g
7	10	ISO 4032 - M5	12.30 g
8	4	ISO 4762 - M2 x 16	1.85 g
9	4	ISO 4762 - M3 x 12	4.20 g
10	4	ISO 4762 - M3 x 16	5.00 g
11	8	ISO 4762 - M4 x 12	15.60 g
12	5	ISO 4762 - M4 x 16	11.50 g
13	2	ISO 4762 - M4 x 8	3.20 g
14	4	ISO 4762 - M5 x 16	15.20 g
15	14	ISO 4762 - M5 x 20	61.60 g
16	1	KP08 Block Bearing	33.00 g
17	2	Limit Switch	5.00 g
18	2	Limit Switch Spacer	1.12 g
19	2	Linear Bearing Spacer	3.26 g
20	2	Linear Guide Rods	145.60 g
21	4	M2 Washer	0.10 g
22	2	M3 Washer	0.20 g
23	2	M4 Insert Nut	0.84 g
24	8	M5 Insert Nut	10.00 g
25	18	M5 Washer	7.92 g
26	1	Pneumatic Fitting	7.00 g
27	1	Product Base	32.52 g
28	2	SCS8UU Linear Bearing	92.00 g
29	2	SHF8 Guide Holder	33.00 g
30	1	SK12 Shaft Suppot	30.00 g
31	2	SK8 Shaft Support	48.00 g
32	1	Suction Cup	8.00 g
33	1	Suction Cup Fixation	59.48 g
34	1	T8 Nut Housing	26.00 g
35	1	T8 nut	12.00 g
36	1	Tr8x8 Lead Screw	53.32 g
37	4	V-Pulley	74.00 g
38	1	Z Bracket	203.25 g

1. Inertia Calculation (J_{eq})

Variables & Parameters:

- Load Mass: $m_{load} = 1{,}363.87\ \text{g} = 1.36387\ \text{kg}$
- Mechanical Efficiency (assumed): $\eta = 90\% = 0.9$
- Pitch Radius: $r = 9.55\ mm$

Equations & Substitution:

- Load Rotary Inertia (J_{load}):

$$J_{load} = \frac{m_{load} \cdot r^2}{\eta} = \frac{1.36387 \cdot (0.00955)^2}{0.9} = 1.3821 \times 10^{-4}\ \text{kg} \cdot \text{m}^2$$

- Motor Inertia (J_{motor}):

$$J_{motor} = \frac{1}{10} \cdot J_{load} = 0.1 \times 1.3821 \times 10^{-4} = 1.3821 \times 10^{-5}\ \text{kg} \cdot \text{m}^2$$

- Equivalent System Inertia (J_{eq}):

$$J_{eq} = J_{load} + J_{motor} = 1.3821 \times 10^{-4} + 1.3821 \times 10^{-5} = 1.5203 \times 10^{-4}\ \text{kg} \cdot \text{m}^2$$

2. Resisting Torque Calculation (T_r)

Variables & Parameters:

- Load Mass: $m_{load} = 1{,}363.87\ \text{g} = 1.36387\ \text{kg}$
- Mechanical Efficiency (assumed): $\eta = 90\% = 0.9$
- Pitch Radius: $r = 9.55\ mm$

Torque Equations & Substitution:

- Total Resisting Torque (T_r):

$$T_r = \frac{m_{load} \cdot r}{\eta} = \frac{1.36387 \cdot 0.00955}{0.9} = 0.01447\ \text{N} \cdot \text{m} \quad (14.47\ \text{mN} \cdot \text{m})$$

3.Acceleration ($\ddot{\theta}_m$):

The stroke of the y axis is **310 mm** and the desired maximum nut speed is **120 mm/s**, so at constant speed the whole stroke would take about **2.58 s** (310 ÷ 120). The acceleration and deceleration ramps of the velocity profile make the actual move a little longer, as calculated below.

Variables & Parameters:

- Stroke: $S = 310$ mm $= 0.31$ m
- Pitch Radius: $r = 9.55\ mm$
- Maximum Linear Speed (desired): $v_{\max} = 120$ mm/s $= 0.120$ m/s
- Phase Time Ratio (assumed, symmetric profile): $t_{acc}: t_{const}: t_{dec} = 0.3: 0.4: 0.3$ of the move time T
- Dwell Time (assumed): $t_{dw} = 0.2$ s
- No gearbox $\Longrightarrow \theta_m = \theta_L$

Equations & Substitution:

- Maximum Angular Velocity ($\omega_{\max}$):

$$\omega_{\max} = \frac{v_{\max}}{r} = \frac{0.120 \text{ m/s}}{0.00955 \text{ m}} = 12.5654 \text{ rad/s} \approx 120.0 \text{ rpm}$$

- Total Rotation for the 310 mm Stroke (θ):

$$\theta = \frac{S}{r} = \frac{0.310 \text{ m}}{0.00955 \text{ m}} = 32.4607 \text{ rad} \approx 5.166 \text{ rev}$$

- Phase Times, $t_{acc}: t_{const}: t_{dec} = 0.3: 0.4: 0.3$ of the total move time T:

$$\theta = \frac{1}{2}(0.3T)\omega_{\max} + \omega_{\max}(0.4T) + \frac{1}{2}(0.3T)\omega_{\max} = 0.7 \cdot \omega_{\max} \cdot T$$

$$T = \frac{\theta}{0.7 \cdot \omega_{\max}} = \frac{32.4607}{0.7 \cdot 12.5654} = 3.6903 \text{ s}$$

- $t_{acc} = 0.3 \times 3.6903 = \mathbf{1.1071}$ s
- $t_{const} = 0.4 \times 3.6903 = \mathbf{1.4761}$ s
- $t_{dec} = 0.3 \times 3.6903 = \mathbf{1.1071}$ s
- Total Cycle Time ($T + t_{dw}$) = $3.6903 + 0.2 = \mathbf{3.8903}$ s

- Acceleration Phase ($\ddot{\theta}_{acc}$):

$$\ddot{\theta}_{acc} = \frac{\omega_{max} - 0}{t_{acc}} = \frac{12.5654}{1.1071} = +\mathbf{11.350}\ \text{rad/s}^2$$

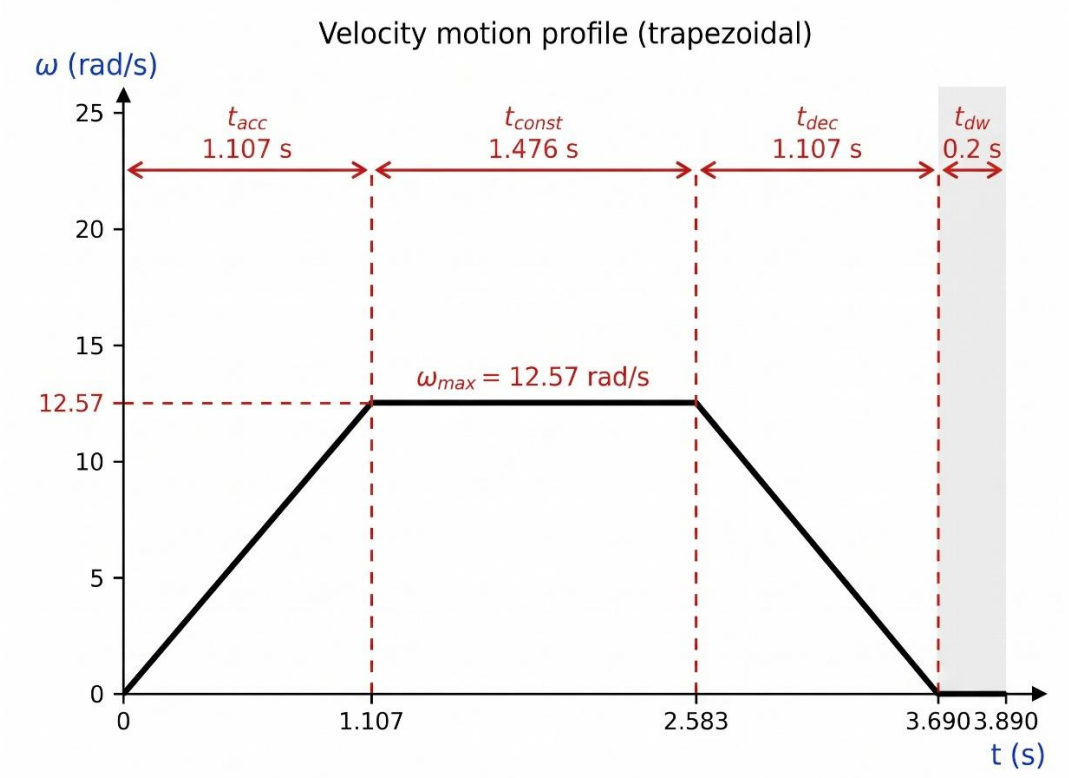
Constant-Speed Phase ($\ddot{\theta}_{const}$):

$$\ddot{\theta}_{const} = 0\ \text{rad/s}^2$$

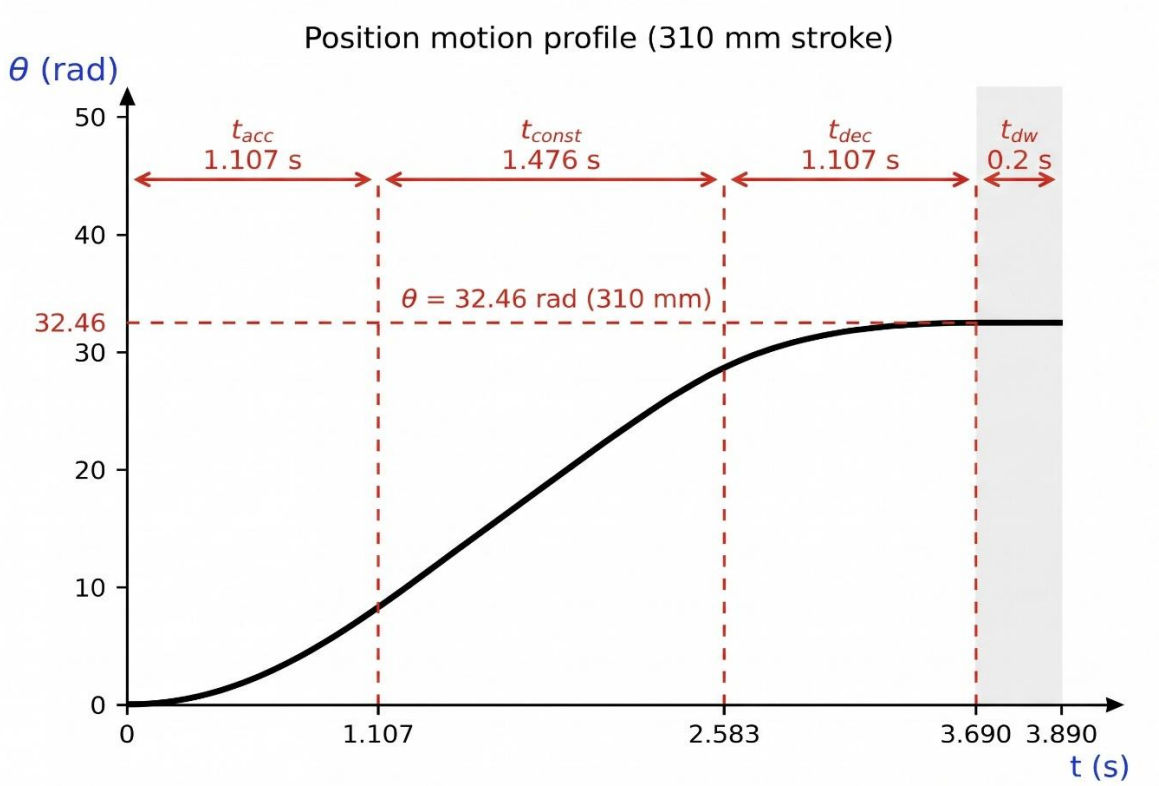
- Deceleration Phase ($\ddot{\theta}_{dec}$):

$$\ddot{\theta}_{dec} = \frac{0 - \omega_{max}}{t_{dec}} = -\mathbf{11.350}\ \text{rad/s}^2$$

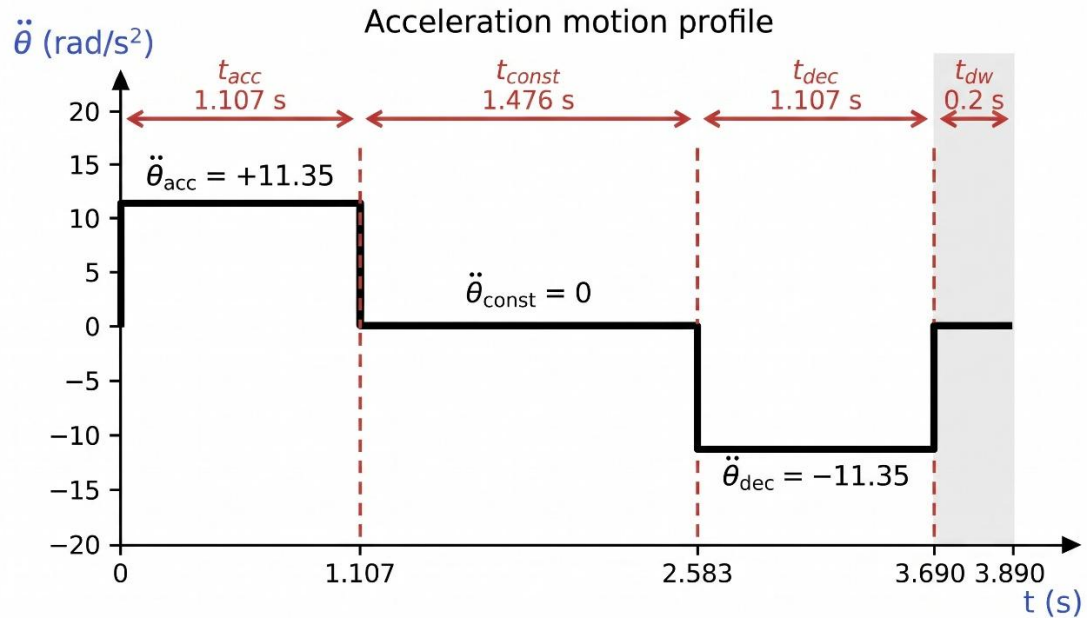
- Velocity Motion Profile (ω vs t):



- Position Motion Profile (θ vs t):



- Acceleration Motion Profile ($\ddot{\theta}$ vs t):



4. Motor Torque Calculations (T_m):

Variables & Parameters:

- Equivalent System Inertia: $J_{\text{eq}} = 1.5203 \times 10^{-4}$ kg · m^2
- Total Resisting Torque: $T_r = 0.01447$ N · m
- Acceleration Phase Angular Acceleration: $\ddot{\theta}_{\text{acc}} = +11.350$ rad/s^2
- Constant-Speed Phase Angular Acceleration: $\ddot{\theta}_{\text{const}} = 0$ rad/s^2
- Deceleration Phase Angular Acceleration: $\ddot{\theta}_{\text{dec}} = -11.350$ rad/s^2
- Phase Times: $t_{\text{acc}} = 1.1071$ s, $t_{\text{const}} = 1.4761$ s, $t_{\text{dec}} = 1.1071$ s, $t_{\text{dw}} = 0.2$ s

Torque Calculations & Substitution:

1. Acceleration Phase Motor Torque ($T_{m,\text{acc}}$):

$$T_{m,\text{acc}} = T_r + J_{\text{eq}} \cdot \ddot{\theta}_{\text{acc}}$$

$$T_{m,\text{acc}} = 0.01447 + (1.5203 \times 10^{-4} \cdot 11.350)$$

$$T_{m,\text{acc}} = 0.01447 + 0.001726 = 0.01620 \text{ N} \cdot \text{m} \quad (16.20 \text{ mN} \cdot \text{m})$$

2. Constant-Speed Phase Motor Torque ($T_{m,\text{const}}$):

$$T_{m,\text{const}} = T_r + J_{\text{eq}} \cdot \ddot{\theta}_{\text{const}}$$

$$T_{m,\text{const}} = 0.01447 + (1.5203 \times 10^{-4} \cdot 0) = 0.01447 \text{ N} \cdot \text{m} \quad (14.47 \text{ mN} \cdot \text{m})$$

3. Deceleration Phase Motor Torque ($T_{m,\text{dec}}$):

$$T_{m,\text{dec}} = T_r + J_{\text{eq}} \cdot \ddot{\theta}_{\text{dec}}$$

$$T_{m,\text{dec}} = 0.01447 + (1.5203 \times 10^{-4} \cdot -11.350)$$

$$T_{m,\text{dec}} = 0.01447 - 0.001726 = 0.01274 \text{ N} \cdot \text{m} \quad (12.74 \text{ mN} \cdot \text{m})$$

4. Dwell Phase Motor Torque ($T_{m,\text{dw}}$):

$$T_{m,\text{dw}} = T_{m,\text{const}} = 0.01447 \text{ N} \cdot \text{m}$$

Peak and RMS Torque Selection:

- Peak Motor Torque ($T_{m,\text{peak}}$):

$$T_{m,\text{peak}} = \max\left(T_{m,\text{acc}}, T_{m,\text{const}}, T_{m,\text{dec}}\right)$$

$$T_{m,\text{peak}} = 0.01620 \text{ N} \cdot \text{m} \quad (16.20 \text{ mN} \cdot \text{m})$$

- Root Mean Square Torque (T_{rms}):

$$T_{\text{rms}} = \sqrt{\frac{T_{m,\text{acc}}^2 \cdot t_{\text{acc}} + T_{m,\text{const}}^2 \cdot t_{\text{const}} + T_{m,\text{dec}}^2 \cdot t_{\text{dec}} + T_{m,\text{dw}}^2 \cdot t_{\text{dw}}}{t_{\text{acc}} + t_{\text{const}} + t_{\text{dec}} + t_{\text{dw}}}}$$

$$T_{\text{rms}} = \sqrt{\frac{(0.01620)^2 \cdot 1.1071 + (0.01447)^2 \cdot 1.4761 + (0.01274)^2 \cdot 1.1071 + (0.01447)^2 \cdot 0.2}{3.8903}}$$

$$T_{\text{rms}} = \sqrt{\frac{0.0002905 + 0.0003091 + 0.0001797 + 0.00004188}{3.8903}} = \sqrt{\frac{0.00082118}{3.8903}}$$

$$= \sqrt{0.00021108}$$

$$T_{\text{rms}} = 0.01453 \text{ N} \cdot \text{m}$$

5.Motor Selection & Verification with Safety Factor

- Safety Factor Application **($n = 4$):**
  Applying a design safety factor of $n = 4$ ensures extra margin for friction, damping and dynamic load spikes:

  $T_{\text{req, rms}} = T_{m,\text{rms}} \times n = 0.01453\ \text{N} \cdot \text{m} \times 4 = \mathbf{0.05812}\ \text{N} \cdot \text{m} \quad (0.5924\ \text{kg} \cdot \text{cm})$

- Selected DC Motor: GA25-370 DC Motor with Encoder 12V 250RPM



Model	JGA25-370
Operating Voltage	12V – 18V DC
Rated Voltage	12V DC
No-Load Speed	250 RPM
No-Load Current	~50 mA
Rated Torque	4.3 kg·cm (0.422 N·m)
Stall Torque	9.0 kg·cm (0.883 N·m)
Stall Current	~1.2 A
Gear Reduction Ratio	1:34
Feedback	Incremental Encoder (Hall Effect)

Suitability Justification: The selected **JGA25-370 DC Motor** easily handles the maximum dynamic load requirement during motion. Since the 20x20 Extrusion beam supports the full 1.36 kg weight, the motor only needs to overcome a small resisting torque of $0.0145\ \text{N} \cdot \text{m}$.

In addition, the motor's no-load speed is 250 RPM, while the maximum required speed is only 120 RPM, which leaves a comfortable speed margin (the motor runs at roughly 48% of its no-load speed). The motor is therefore suitable for this application.

Note: Even after updating the system inertia with the actual motor value, the recalculated peak required torque with a safety factor of $n = 4$ is only $0.0653\ \text{N} \cdot \text{m}$ ($0.666\ \text{kg} \cdot \text{cm}$). The motor's rated torque of $4.3\ \text{kg} \cdot \text{cm}$ **(**$0.4217\ \text{N} \cdot \text{m}$**)** is still nearly $6.5\times$ **higher** than this requirement, confirming that the JGA25-370 remains capable of driving the load with a safety margin.